
Experiment 04

Heat Transfer in Turbulent Flow



Chemical Engineering Laboratory Report

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1 Aim

To determine the overall heat-transfer coefficient of a double-pipe heat exchanger by the logarithmic mean temperature difference method, estimate the individual film coefficients using a Wilson plot, and compare the tube-side coefficient with the Dittus–Boelter correlation for turbulent flow.

2 Apparatus

- Double-pipe heat exchanger test section.
- Hot-water reservoir fitted with an electric heater and temperature controller.
- Cold-water reservoir and condenser.
- Separate circulation pumps and rotameters for the hot and cold streams.
- Digital temperature display connected to the four inlet and outlet sensors.
- 230 V A.C. electrical supply with 15 A and 5 A connections.

3 Procedure

1. The 15 A and 5 A plugs were connected to the 230 V A.C. supply.
2. The heater controller was set within 65–75°C, and the apparatus was allowed to approach steady operation.
3. The cold-water circuit was started and its flow rate was fixed at the maximum recorded setting of 420 L min⁻¹ for the entire run.
4. The hot-water circulation pump was switched on and the rotameter was adjusted to the first flow rate.
5. After the displayed temperatures became steady, the hot inlet and outlet temperatures (T_1, T_2) and the cold inlet and outlet temperatures (t_1, t_2) were recorded.
6. The procedure was repeated for six hot-water flow rates from 120 to 420 L h⁻¹.

4 Experimental Setup

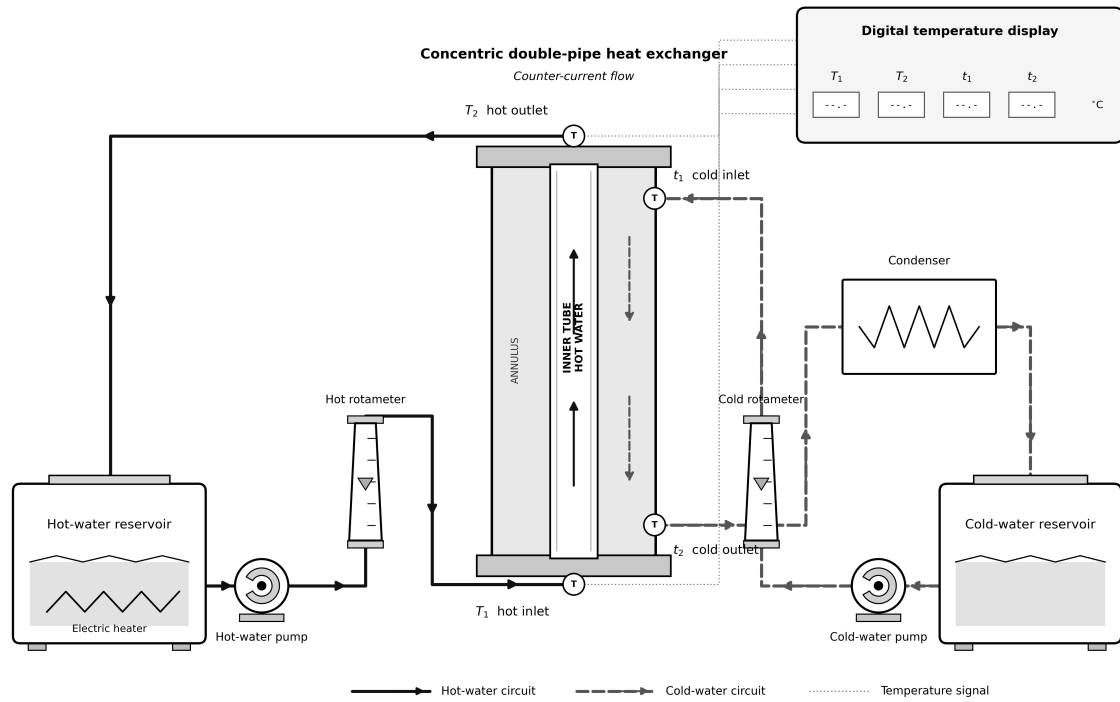


Figure 1: Schematic of the turbulent-flow heat-transfer apparatus. The hot stream passes through the inner tube and the cold stream through the annulus in counter-current flow.

The hot-water reservoir, heater, pump and rotameter form the tube-side circuit. The cold-water reservoir, pump, rotameter and condenser form the annulus-side circuit. The four temperature sensors are connected to the digital display. Solid and dashed lines distinguish the two circulating streams in Figure 1.

5 Theory

Heat is transferred from the hot water to the cold water through the wall of the inner tube. Turbulent mixing reduces the thermal boundary-layer resistance and generally raises the convective heat-transfer coefficient.

5.1 Overall heat-transfer coefficient

For the inside film, tube wall and outside film acting as resistances in series,

$$\frac{1}{U_i A_i} = \frac{1}{h_i A_i} + \frac{x}{K A_i} + \frac{1}{h_o A_o}. \quad (1)$$

Following the laboratory-manual treatment, the thin-wall resistance and the area difference are neglected. Hence,

$$\frac{1}{U_i} = \frac{1}{h_i} + \frac{1}{h_o}. \quad (2)$$

5.2 Logarithmic mean temperature difference

For counter-current operation,

$$\Delta T_{\text{lm}} = \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln \left[\frac{T_1 - t_2}{T_2 - t_1} \right]}. \quad (3)$$

The hot-side heat duty and the overall coefficient are

$$\dot{Q} = \dot{m}C_p(T_1 - T_2), \quad U_i = \frac{\dot{Q}}{A_i \Delta T_{\text{lm}}}. \quad (4)$$

5.3 Wilson plot

The cold-stream flow rate was kept constant, so h_o was treated as constant. For turbulent tube flow, $h_i \propto V^{0.8}$; therefore,

$$\frac{1}{U_i} = C \left(\frac{1}{V^{0.8}} \right) + \frac{1}{h_o}. \quad (5)$$

Thus, the intercept of a plot of $1/U_i$ against $1/V^{0.8}$ gives $1/h_o$. In this report, V is expressed in m s^{-1} .

5.4 Dittus–Boelter correlation

For fully developed turbulent flow through a smooth tube,

$$Re = \frac{\rho V d_1}{\mu}, \quad Pr = \frac{C_p \mu}{K}, \quad Nu = 0.023 Re^{0.8} Pr^{0.3}, \quad (6)$$

where the exponent 0.3 is used because the hot stream is being cooled. The predicted inside film coefficient is

$$h_{i,\text{theo}} = \frac{NuK}{d_1}. \quad (7)$$

6 Observations

Table 1: Physical constants and exchanger dimensions used in the calculations.

Quantity	Value
Heat-exchanger length, L	1.0 m
Inside diameter of inner tube, d_1	0.007 m
Outside diameter of inner tube, d_2	0.010 m
Inside diameter of outer tube, D	0.018 m
Specific heat capacity of water, C_p	1.0 kcal kg ⁻¹ °C ⁻¹
Density of water, ρ	1000 kg m ⁻³
Thermal conductivity of water, K	0.5 kcal h ⁻¹ m ⁻¹ °C ⁻¹
Dynamic viscosity of water, μ	0.47 cP = 4.7×10^{-4} kg m ⁻¹ s ⁻¹
Cold-water flow rate, held constant	420 L min ⁻¹
Inner heat-transfer area, $A_i = \pi d_1 L$	0.021991 m ²
Inner flow area, $A_c = \pi d_1^2/4$	3.84845×10^{-5} m ²
Prandtl number, Pr	3.384

Table 2: Recorded temperatures at each hot-water flow rate.

Sr. No.	T_1 ($^{\circ}\text{C}$)	T_2 ($^{\circ}\text{C}$)	t_1 ($^{\circ}\text{C}$)	t_2 ($^{\circ}\text{C}$)	Hot flow (L/h)
1	65.4	51.9	30.0	33.5	120
2	64.5	54.6	30.6	34.8	180
3	65.0	56.9	31.7	36.3	240
4	65.3	58.2	32.2	37.2	300
5	65.2	58.7	32.8	37.9	360
6	65.2	59.5	33.2	38.5	420

7 Calculations

7.1 Sample calculation for run 3

For a hot-water flow rate of 240 L h^{-1} ,

$$\dot{m} = 240 \text{ kg h}^{-1}, \quad \dot{Q} = 240(1.0)(65.0 - 56.9) = 1944.0 \text{ kcal h}^{-1}.$$

The terminal temperature differences are

$$\Delta T_1 = 65.0 - 36.3 = 28.7^{\circ}\text{C}, \quad \Delta T_2 = 56.9 - 31.7 = 25.2^{\circ}\text{C}.$$

Therefore,

$$\Delta T_{\text{lm}} = \frac{28.7 - 25.2}{\ln(28.7/25.2)} = 26.91^{\circ}\text{C},$$

$$U_i = \frac{1944.0}{(0.021991)(26.9121)} = 3284.74 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^{\circ}\text{C}^{-1}, \quad \frac{1}{U_i} = 3.04438 \times 10^{-4}.$$

The mean tube velocity is

$$V = \frac{240 \times 10^{-3}/3600}{3.84845 \times 10^{-5}} = 1.7323 \text{ m s}^{-1}, \quad \frac{1}{V^{0.8}} = 0.64432.$$

The fitted Wilson relation obtained from all six runs is

$$\boxed{\frac{1}{U_i} = (1.68749 \times 10^{-4}) \left(\frac{1}{V^{0.8}} \right) + 1.82332 \times 10^{-4}}, \quad R^2 = 0.9398. \quad (8)$$

Hence,

$$h_o = \frac{1}{1.82332 \times 10^{-4}} = 5484.50 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^{\circ}\text{C}^{-1}.$$

For run 3,

$$\frac{1}{h_{i,\text{calc}}} = \frac{1}{U_i} - \frac{1}{h_o} = 3.04438 \times 10^{-4} - 1.82332 \times 10^{-4},$$

$$h_{i,\text{calc}} = 8189.60 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^{\circ}\text{C}^{-1}.$$

Also,

$$Re = \frac{1000(1.7323)(0.007)}{4.7 \times 10^{-4}} = 25800.19,$$

$$Nu = 0.023(25800.19)^{0.8}(3.384)^{0.3} = 112.16,$$

$$h_{i,\text{theo}} = \frac{112.16(0.5)}{0.007} = 8011.75 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^{\circ}\text{C}^{-1}.$$

The signed percentage deviation is

$$\% \text{ deviation} = \frac{8189.60 - 8011.75}{8011.75} \times 100 = +2.22\%.$$

8 Calculated Results

Table 3: Heat duty, LMTD, velocity and overall coefficient for each run.

Run	$\dot{Q}$ (kcal/h)	ΔT_{lm} ($^{\circ}\text{C}$)	V (m/s)	$1/V^{0.8}$	U_i ($\text{kcal h}^{-1} \text{m}^{-2} \text{ }^{\circ}\text{C}^{-1}$)	$1/U_i$ ($\times 10^{-4}$)
1	1620.0	26.5873	0.8661	1.12183	2770.72	3.60917
2	1782.0	26.7489	1.2992	0.81106	3029.39	3.30100
3	1944.0	26.9121	1.7323	0.64432	3284.74	3.04438
4	2130.0	27.0364	2.1654	0.53898	3582.47	2.79137
5	2340.0	26.5939	2.5984	0.46583	4001.17	2.49927
6	2394.0	26.4995	3.0315	0.41179	4108.08	2.43423

Table 4: Comparison of the calculated and Dittus–Boelter inside film coefficients.

Re	Pr	Nu	$h_{i,calc}$ ($\text{kcal h}^{-1} \text{m}^{-2} \text{ }^{\circ}\text{C}^{-1}$)	$h_{i,theo}$ ($\text{kcal h}^{-1} \text{m}^{-2} \text{ }^{\circ}\text{C}^{-1}$)	Deviation (%)
12900.10	3.384	64.42	5599.58	4601.54	+21.69
19350.15	3.384	89.11	6767.37	6364.68	+6.33
25800.19	3.384	112.16	8189.60	8011.75	+2.22
32250.24	3.384	134.09	10330.05	9577.57	+7.86
38700.29	3.384	155.14	14793.95	11081.54	+33.50
45150.34	3.384	175.50	16369.04	12535.96	+30.58
Mean absolute percentage deviation					17.03

9 Wilson Plot

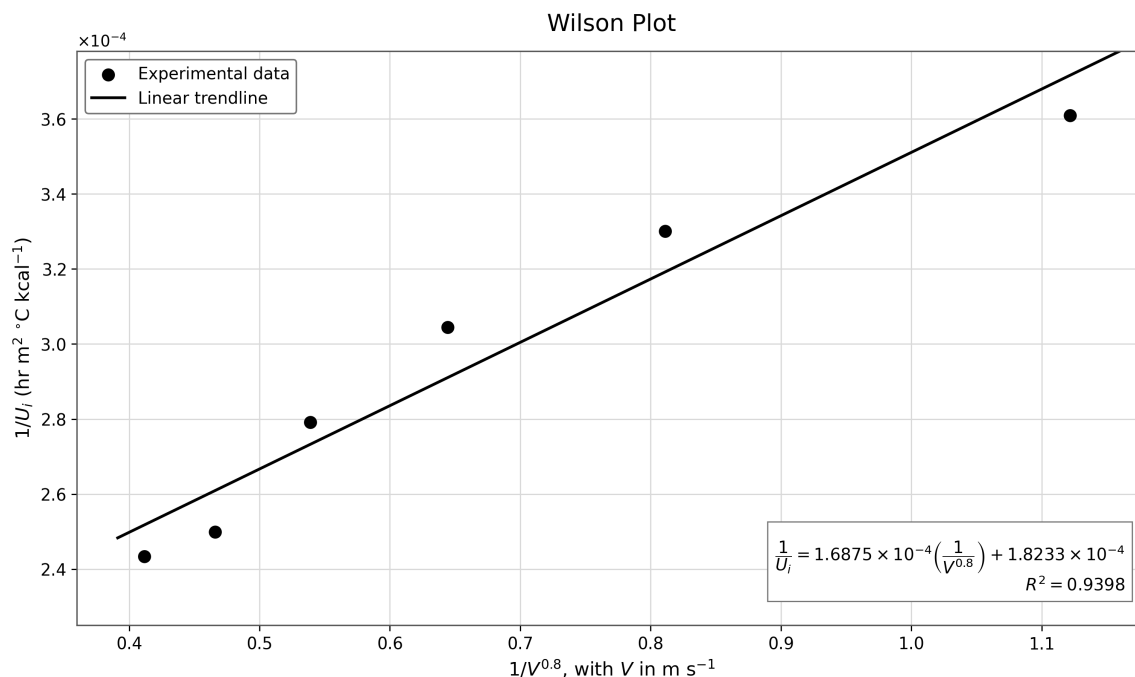


Figure 2: Wilson plot obtained from the six experimental runs. The intercept of the least-squares line gives $1/h_o$.

Figure 2 was generated directly from the full-precision values in the laboratory Excel workbook. The graph uses V in m s⁻¹, consistent with Table 3 and Equation 8.

10 Discussion

- **Wilson-plot linearity:** The six points follow a reasonably strong linear relation, with $R^2 = 0.9398$. This supports the assumed dependence $h_i \propto V^{0.8}$ over the tested range. The remaining scatter is greatest at the higher-flow settings.
- **Effect of hot-water flow rate:** As the flow rate increased from 120 to 420 L h⁻¹, the Reynolds number rose from approximately 1.29×10^4 to 4.52×10^4 . Over the same interval, U_i increased from 2770.72 to 4108.08 kcal h⁻¹ m⁻² °C⁻¹. The increase is consistent with improved turbulent mixing and a thinner tube-side thermal boundary layer.
- **Comparison with Dittus–Boelter:** The calculated and theoretical values of h_i show the same increasing trend. Agreement is closest for the 180–300 L h⁻¹ runs, where the deviation is about 2–8%. The two highest-flow runs deviate by about 31–34%, raising the mean absolute percentage deviation to 17.03%.

11 Error Analysis

- Heat lost from the hot-fluid circuit to the surroundings is included in the hot-side temperature drop but is not transferred entirely to the cold stream. This can make the calculated exchanger duty and U_i higher than their actual values.
- Water properties were treated as constant even though the bulk temperature changed across the exchanger.

- Rotameter parallax and flow fluctuations directly affect V , Re and the horizontal coordinate of the Wilson plot.
- Since U_i is inversely proportional to LMTD, small uncertainty in any of the four temperature readings is amplified in the calculated coefficient.
- Heater cycling and incomplete steady-state operation may have introduced temperature fluctuations, particularly at the highest flow rates.
- With only six runs, the regression intercept and the resulting value of h_o remain sensitive to individual points.

12 Assumptions

1. Flow through the inner tube was fully developed and turbulent.
2. The apparatus operated at steady state while each set of readings was recorded.
3. Water properties were constant over the experimental temperature range.
4. Tube-wall and fouling resistances were negligible in comparison with the film resistances.
5. The annulus-side coefficient remained constant because the cold-water flow rate was fixed.
6. Radiative heat transfer, entrance effects and axial heat conduction were negligible.

13 Applications

- Design and rating of double-pipe and shell-and-tube heat exchangers.
- Heating and cooling duties in refineries and chemical-process plants.
- Waste-heat recovery, feed-water heating and power-generation systems.
- Pasteurisation, sterilisation and controlled cooling in food and pharmaceutical processing.
- Condensers, evaporators and refrigeration or HVAC equipment.

14 Conclusion

The overall heat-transfer coefficient increased from 2770.72 to 4108.08 kcal h⁻¹ m⁻² °C⁻¹ as the hot-water flow rate rose from 120 to 420 L h⁻¹. The Wilson plot was reasonably linear ($R^2 = 0.9398$) and gave an annulus-side film coefficient of $h_o = 5484.50$ kcal h⁻¹ m⁻² °C⁻¹ under the assumptions used. The calculated tube-side coefficients followed the Dittus–Boelter trend, with a mean absolute percentage deviation of 17.03%. The larger differences at the two highest flow rates are consistent with the observed Wilson-plot scatter and the sensitivity of U_i to flow-rate and temperature measurement errors.

15 References

1. CL 326 Integrated Chemical Engineering Laboratory – I manual, Experiment 04: Heat Transfer in Turbulent Flow, IIT Gandhinagar.
2. Experimental observations and calculation workbook prepared during the laboratory session on 08 September 2026.

16 Acknowledgement

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